Course 02323/02402 Introduction to Statistics

Lecture 8: Simple linear regression

DTU Compute
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2800 Lyngby – Denmark

Overview

1. Summary
2. Example: Linear regression with height and weight data
3. Linear regression models
4. Least squares method
5. Statistics and linear regression
   - Matrix notation
6. Hypothesis tests and confidence intervals for β_0 and β_1
7. Confidence and prediction intervals for the line
8. Correlation and Explained Variance
9. Model validation - Analysis of residuals

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Summary



- For two groups, the explanatory variable is “just” which group you belong to.
- Now we will talk about **continuous** explanatory variables.

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Example: Linear regression with height and weight data

- What should we be able to do after today?
- Go to today's Python notebook in VS Code
  - "Example: Height and weight"

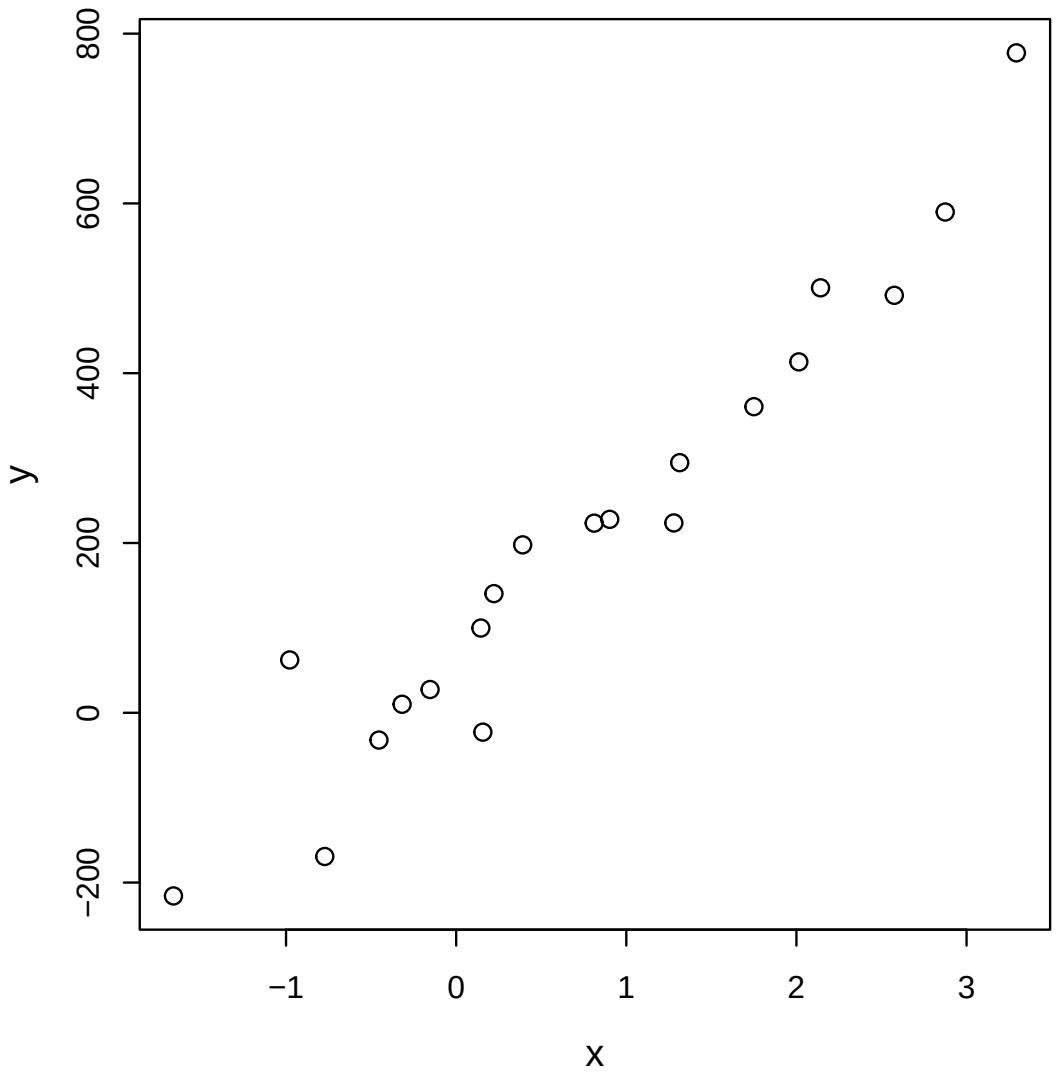


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Start with a scatterplot

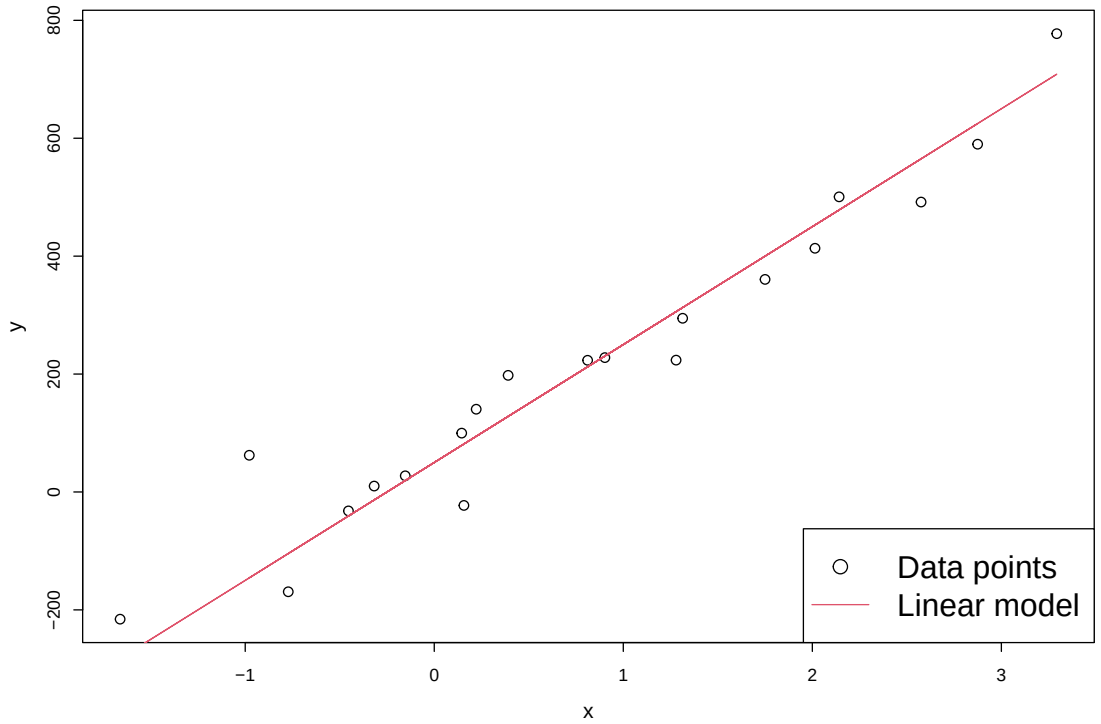
- We have n pairs of data points (x_i, y_i).



A linear model

If the data points lie on a straight line, the relationship between x and y values can be described by the equation:

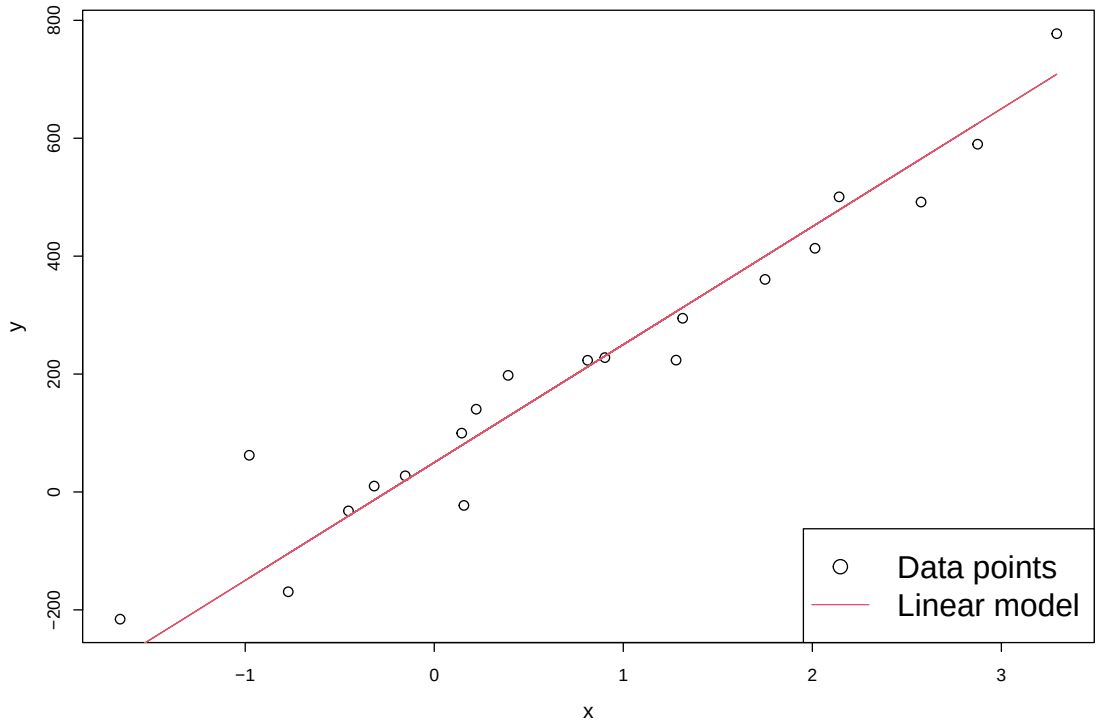
$$y_i = \beta_0 + \beta_1 x_i.$$



A linear model

If the data points lie on a straight line, the relationship between x and y values can be described by the equation:

$$y_i = \beta_0 + \beta_1 x_i.$$



- We need a description of the *random variation*.

The linear regression model

- The *linear regression model*:

$$Y_i = \beta_0 + \beta_1 x_i + \varepsilon_i \quad (i = 1, \ldots, n).$$

- Y_i is the *dependent variable*.
- x_i is the *explanatory variable*.
- ε_i is the deviation (residual).
- We assume $\varepsilon_i \sim N(0, \sigma^2)$ (and i.i.d.).

The linear regression model

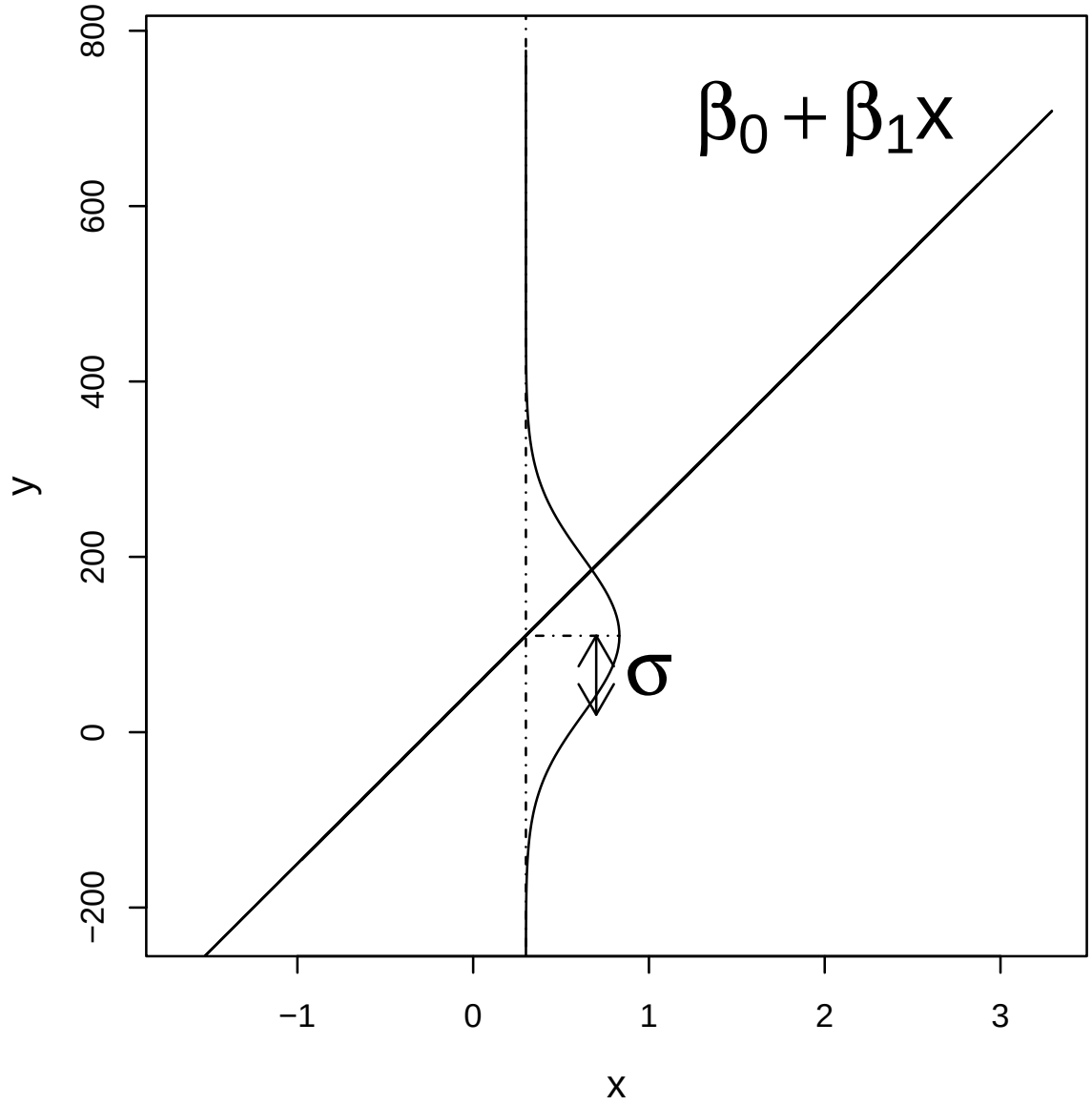
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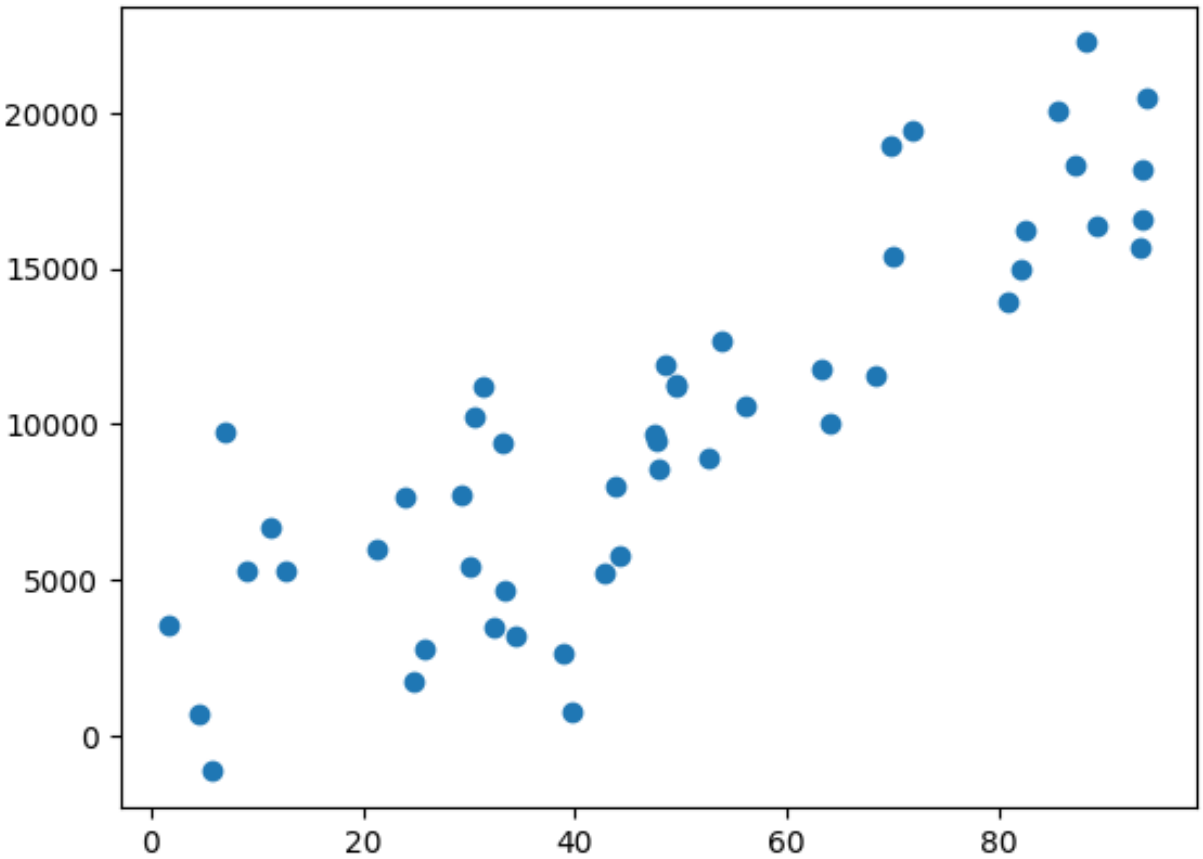
- Y_i is the *dependent variable*.
- x_i is the *explanatory variable*.
- ε_i is the deviation (residual).
- We assume $\varepsilon_i \sim N(0, \sigma^2)$ (and i.i.d.).

Consider: *What kind of distribution does Y_i follow? Are the Y_is identically distributed?*

Illustration of the statistical model



Kahoot



Kahoot!

(x7)

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Least squares method

- We want to estimate the parameters β_0 and β_1.

Least squares method

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- Good idea: Let's minimize the variance of the residuals (σ^2).

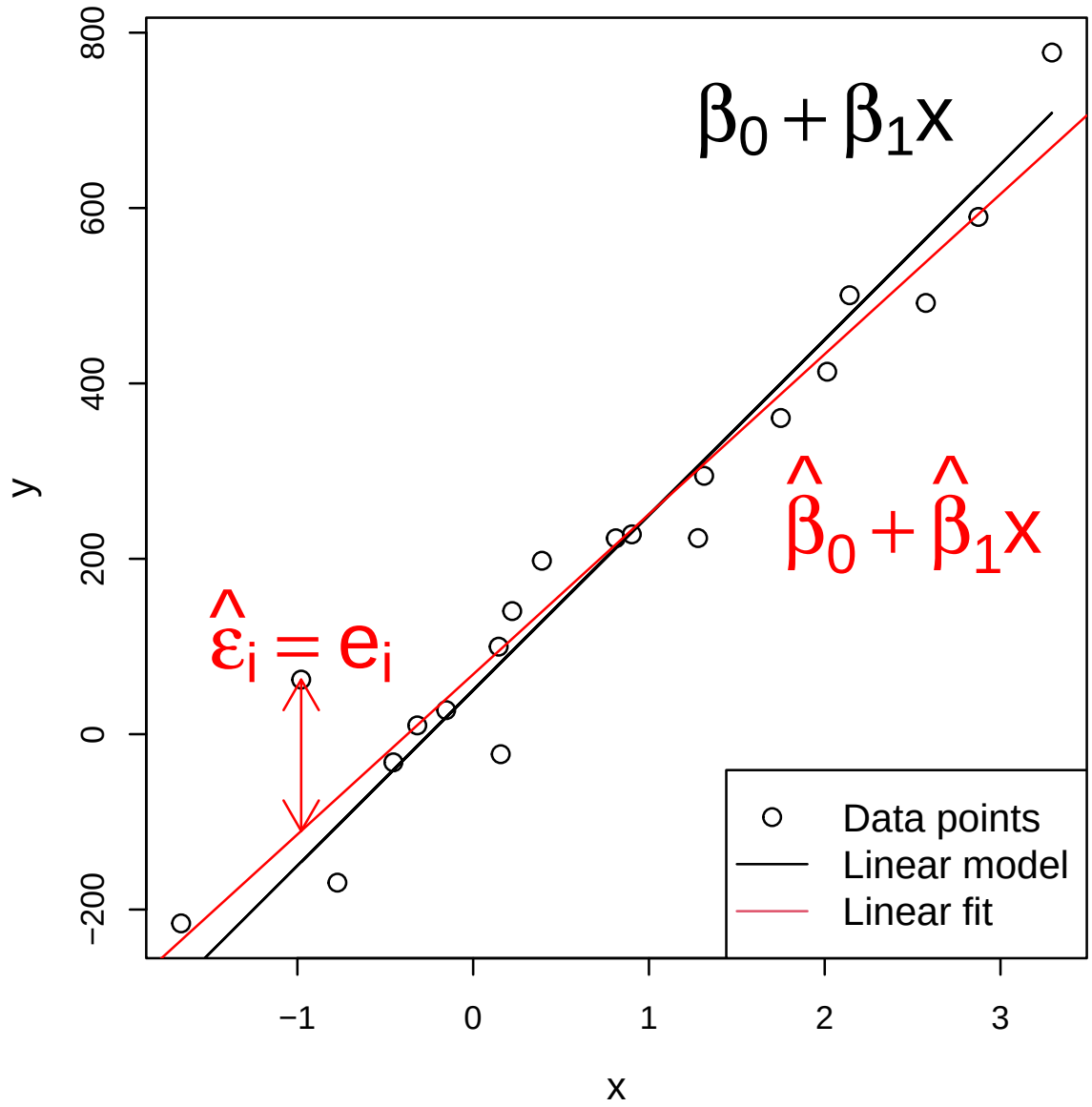
Least squares method

- We want to estimate the parameters β_0 and β_1.
- Good idea: Let's minimize the variance of the residuals (σ^2).
- We minimize the sum of the squared residuals (Residual Sum of Squares, RSS):

$$RSS(\beta_0, \beta_1) = \sum_{i=1}^{n} \varepsilon_i^2 = \sum_{i=1}^{n} (y_i - \beta_0 - \beta_1 x_i)^2 .$$

That is, we choose $\hat{\beta}_0$ and $\hat{\beta}_1$ so that they minimize RSS.

Illustration of model, data, and fit



'Least squares' estimators

Theorem 5.4 (for estimators, as in the book)

'Least squares' estimators for β_0 and β_1 are given by:

$$\hat{\beta}_1 = \frac{\sum_{i=1}^{n}(Y_i - \bar{Y})(x_i - \bar{x})}{S_{xx}},$$

$$\hat{\beta}_0 = \bar{Y} - \hat{\beta}_1 \bar{x},$$

where $S_{xx} = \sum_{i=1}^{n}(x_i - \bar{x})^2$.

'Least squares' estimates

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where $S_{xx} = \sum_{i=1}^{n}(x_i - \bar{x})^2$.

Example in Python: Estimate β_0 and β_1

- Go to today's Python notebook in VS Code
  - Now we will try to simulate some data from a known underlying model. Based on the data, we will estimate the parameters.
  - "Example: Estimating parameters β_0 and β_1"



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Simulation: Variation in parameter estimates

We take a new sample

Will the estimates of $\hat{\beta}_0$ and $\hat{\beta}_1$ be the same?

- Go to today's Python notebook in VS Code
  - "Example: Variation of β_0 and β_1"



Thinking break

What did we learn from the simulation?

What are the distributions of $\hat{\beta}_0$ and $\hat{\beta}_1$?

What (do you think) affects the variance of $\hat{\beta}_0$ and $\hat{\beta}_1$?

(Write your own notes here)

Simulation: Variation in parameter estimates

There is variation in the parameters!

A new sample leads to new realizations of the estimators, i.e., new estimates.

Simulation: Variation in parameter estimates

There is variation in the parameters!

A new sample leads to new realizations of the estimators, i.e., new estimates.

What are the distributions of the parameter estimators?

We need to know them to create confidence intervals, etc.

Variance/distribution of $\hat{\beta}_0$ and $\hat{\beta}_1$

Theorem 5.8 Variance of estimators

The variance and covariance of the estimators in Theorem 5.4 are given by

$$V[\hat{\beta}_0] = \frac{\sigma^2}{n} + \frac{\bar{x}^2\sigma^2}{S_{xx}}, \tag{5-27}$$

$$V[\hat{\beta}_1] = \frac{\sigma^2}{S_{xx}}, \tag{5-28}$$

$$\mathrm{Cov}[\hat{\beta}_0, \hat{\beta}_1] = -\frac{\bar{x}\sigma^2}{S_{xx}}, \tag{5-29}$$

where σ^2 is usually replaced by its estimate ($\hat{\sigma}^2$). The central estimator for σ^2 is

$$\hat{\sigma}^2 = \frac{RSS(\hat{\beta}_0, \hat{\beta}_1)}{n-2}. \tag{5-30}$$

When the estimate of σ^2 is used the variances also become estimates and we'll refer to them as $\hat{\sigma}^2_{\hat{\beta}_0}$ and $\hat{\sigma}^2_{\hat{\beta}_1}$.

"Standard Errors" for $\hat{\beta}_0$ and $\hat{\beta}_1$

$$\hat{\sigma}_{\beta_0} = \sqrt{\frac{\hat{\sigma}^2}{n} + \frac{\bar{x}^2\hat{\sigma}^2}{S_{xx}}} = \hat{\sigma}\sqrt{\frac{1}{n} + \frac{\bar{x}^2}{\sum_{i=1}^{n}(x_i - \bar{x})^2}}, \tag{5-43}$$

$$\hat{\sigma}_{\beta_1} = \sqrt{\frac{\hat{\sigma}^2}{S_{xx}}} = \hat{\sigma}\sqrt{\frac{1}{\sum_{i=1}^{n}(x_i - \bar{x})^2}}, \tag{5-44}$$

$\hat{\sigma}_{\beta_0}$ and $\hat{\sigma}_{\beta_1}$ are the "**Standard Errors**" for $\hat{\beta}_0$ and $\hat{\beta}_1$!

Furthermore, we have:

$$\hat{\sigma} = \sqrt{\frac{RSS(\hat{\beta}_0,\hat{\beta}_1)}{n-2}} = \sqrt{\frac{\sum_i^n \hat{\varepsilon}_i^2}{n-2}}$$

(know the difference between the various σ's!)

Kahoot!
(x2)

Python

Python function to estimate a linear regression model:

- `my_fit = smf.ols(formula = 'y ~ x', data=...).fit()`

Print the model in a table:

- `print(my_fit.summary(slim=True))`

- Go to today's Python notebook in VS Code
  - "Example: Estimate standard errors of the parameters (height and weight data))"



The General Linear Model for Simple Linear Regression

Model for "Simple Linear Regression":

$$Y_i = \beta_0 + \beta_1 \cdot x_i + \varepsilon_i$$

Can also be written in matrix form:

$$\mathbf{Y} = \mathbf{X}\beta + \varepsilon, \quad \varepsilon_i \sim N(0, \sigma^2)$$

$$\mathbf{Y} = \begin{bmatrix} Y_1 \\ Y_2 \\ \vdots \\ Y_n \end{bmatrix} \qquad \mathbf{X} = \begin{bmatrix} 1 & x_1 \\ 1 & x_2 \\ \vdots & \vdots \\ 1 & x_n \end{bmatrix} \qquad \beta = \begin{bmatrix} \beta_0 \\ \beta_1 \end{bmatrix} \qquad \varepsilon = \begin{bmatrix} \varepsilon_1 \\ \varepsilon_2 \\ \vdots \\ \varepsilon_n \end{bmatrix}$$

Here $\{x_1, x_2, ..., x_n\}$ is a continuous explanatory variable

Theorem 5.23

The estimators of the parameters in the simple linear regression model are given by

$$\hat{\boldsymbol{\beta}} = (\boldsymbol{X}^T\boldsymbol{X})^{-1}\boldsymbol{X}^T\boldsymbol{Y}, \tag{5-67}$$

and the covariance matrix of the estimates is

$$\mathrm{V}[\hat{\boldsymbol{\beta}}] = \sigma^2(\boldsymbol{X}^T\boldsymbol{X})^{-1}, \tag{5-68}$$

and central estimate for the error variance is

$$\hat{\sigma}^2 = \frac{RSS}{n-2}. \tag{5-69}$$

$$\hat{\beta} = \begin{bmatrix} \hat{\beta}_0 \\ \hat{\beta}_1 \end{bmatrix} \qquad \mathbf{V}[\hat{\beta}] = \begin{bmatrix} \mathbf{V}[\hat{\beta}_0] & \mathbf{Cov}[\hat{\beta}_0, \hat{\beta}_1] \\ \mathbf{Cov}[\hat{\beta}_0, \hat{\beta}_1] & \mathbf{V}[\hat{\beta}_1] \end{bmatrix}$$

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Hypothesis tests for β_0 and β_1

We can conduct hypothesis tests for the parameters in a linear regression model:

$$\begin{aligned} H_{0,i}: &\quad \beta_i = \beta_{0,i}, \\ H_{1,i}: &\quad \beta_i \neq \beta_{0,i}. \end{aligned}$$

Hypothesis tests for β_0 and β_1

We can conduct hypothesis tests for the parameters in a linear regression model:

$$H_{0,i}: \quad \beta_i = \beta_{0,i},$$
$$H_{1,i}: \quad \beta_i \neq \beta_{0,i}.$$

Theorem 5.12

Under the null hypotheses ($\beta_0 = \beta_{0,0}$ and $\beta_1 = \beta_{0,1}$), the test statistics are

$$T_{\beta_0} = \frac{\hat{\beta}_0 - \beta_{0,0}}{\hat{\sigma}_{\beta_0}}, \quad T_{\beta_1} = \frac{\hat{\beta}_1 - \beta_{0,1}}{\hat{\sigma}_{\beta_1}},$$

t-distributed with $n-2$ degrees of freedom.

Hypothesis test for β_0 and β_1

- See example 5.13 for an example of a hypothesis test.
- Test whether the parameters are significantly different from 0:

$$H_{0,i} : \beta_i = 0, \quad H_{1,i} : \beta_i \neq 0.$$

- Go to today's Python notebook in VS C
  - "Example: Hypothesis test for parameters"



Confidence intervals for β_0 and β_1

Method 5.15 Parameter confidence intervals

$(1-\alpha)$ confidence intervals for β_0 and β_1 are given by

$$\hat{\beta}_0 \pm t_{1-\alpha/2} \cdot \hat{\sigma}_{\beta_0}, \tag{5-52}$$

$$\hat{\beta}_1 \pm t_{1-\alpha/2} \cdot \hat{\sigma}_{\beta_1}, \tag{5-53}$$

where $t_{1-\alpha/2}$ is the $(1-\alpha/2)$-quantile of a t-distribution with $n-2$ degrees of freedom. Where $\hat{\sigma}_{\beta_0}$ and $\hat{\sigma}_{\beta_1}$ are calculated from the results in Theorem 5.8, and Equations (5-43) and (5-44).

Confidence Intervals for β_0 and β_1

- In the output table (the regression table), $\hat{\sigma}_{\beta_0}$ and $\hat{\sigma}_{\beta_1}$ can be read under "std err".

- Go to the Python notebook VS Code
  - "Example: Confidence interval for parameters"
  - + KAHOOT (x1)



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Method 5.18: Confidence interval for the regression line

A simple linear regression model can be written as

$$Y \sim N(\mu(x), \sigma^2),$$

where $\mu(x) = \beta_0 + \beta_1 x$.
For $x = x_0$, we can find a confidence interval for $\mu(x_0) = \beta_0 + \beta_1 x_0$.

The $(1-\alpha)$ confidence interval for the regression line at $x = x_0$ (for $\mu(x_0)$) can be found by:

$$(\hat{\beta}_0 + \hat{\beta}_1 x_0) \pm t_{\alpha/2} \cdot \hat{\sigma} \sqrt{\frac{1}{n} + \frac{(x_0 - \bar{x})^2}{S_{xx}}}.$$

(in the book, it is called $\hat{\mu}(x_0) = \hat{Y}(x_{new})$)

Python

Python code to calculate $\hat{\mu}(x_0)$ for new x values: Define the desired x values:

- `x_new = pd.DataFrame('x': ... )`

Calculate $\hat{\mu}(x_0)$ for each x value:

- `my_fit.get_prediction(x_new).summary_frame(alpha=...)`

(The confidence interval is obtained using `mean_ci_lower` and `mean_ci_upper`)

- Go to today's Python notebook in VS Code
  - "Example: Confidence interval for the line"



Method 5.18: Prediction interval for a new observation

- We want a prediction interval for a new observation Y_0 at $x = x_0$.
- The $(1-\alpha)$ prediction interval for a new observation Y_0 at $x = x_0$ can be found by:

$$(\hat{\beta}_0 + \hat{\beta}_1 x_0) \pm t_{\alpha/2} \cdot \hat{\sigma} \sqrt{1 + \frac{1}{n} + \frac{(x_0 - \bar{x})^2}{S_{xx}}}.$$

- The prediction interval will contain the observed y_0 in $100(1-\alpha)\%$ of cases.
- For fixed α, the prediction interval is larger than the confidence interval.

Python

Python code to calculate the prediction interval for observations (for new x-values):
Define the desired x-values:

- `x_new = pd.DataFrame('x': ... )`
- `my_fit.get_prediction(x_new).summary_frame(alpha=...)`

(this time, use `obs_ci_lower` and `obs_ci_upper`)

- Go to today's Python notebook in VS Code
  - "Example: Prediction interval for the observations"



Confidence and Prediction Intervals in the Book

Method 5.18 Intervals for the line

The (1-α) **confidence interval** for the line $\hat{\beta}_0 + \hat{\beta}_1 x_{\text{new}}$ is

$$\hat{\beta}_0 + \hat{\beta}_1 x_{\text{new}} \pm t_{1-\alpha/2}\hat{\sigma}\sqrt{\frac{1}{n} + \frac{(x_{\text{new}} - \bar{x})^2}{S_{xx}}}, \tag{5-59}$$

and the (1-α) **prediction interval** is

$$\hat{\beta}_0 + \hat{\beta}_1 x_{\text{new}} \pm t_{1-\alpha/2}\hat{\sigma}\sqrt{1 + \frac{1}{n} + \frac{(x_{\text{new}} - \bar{x})^2}{S_{xx}}}, \tag{5-60}$$

where $t_{1-\alpha/2}$ is the $(1-\alpha/2)$-quantile of the t-distribution with $n-2$ degrees of freedom.

Confidence and prediction intervals

- The confidence interval indicates the uncertainty of the *regression line*.
- The prediction interval indicates the uncertainty of a *new observation*.
- For a fixed α, the prediction interval is larger than the confidence interval.
- The prediction interval can never be smaller than the random variation in the data (i.e., that from the error term ε).

Kahoot!
(x2)

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Explained variance and correlation

- The explained variance in a model is R^2 (R-squared).
- Calculated with

$$R^2 = 1 - \frac{\sum_i (y_i - \hat{y}_i)^2}{\sum_i (y_i - \bar{y})^2},$$

  where $\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_i$.

- The proportion of the total variance explained by the model.

Kahoot!

(x2)

Explained variance and correlation

- The correlation ρ is a measure of *linear relationship* between two stochastic variables.
- The estimated (i.e., empirical) correlation satisfies

$$\hat{\rho} = R = \sqrt{R^2}\, sign(\hat{\beta}_1),$$

where $sign(\hat{\beta}_1)$ is -1 for $\hat{\beta}_1 \leq 0$ and 1 for $\hat{\beta}_1 > 0$

Explained variance and correlation

- The correlation ρ is a measure of *linear relationship* between two stochastic variables.
- The estimated (i.e., empirical) correlation satisfies

$$\hat{\rho} = R = \sqrt{R^2}\, sign(\hat{\beta}_1),$$

  where $sign(\hat{\beta}_1)$ is -1 for $\hat{\beta}_1 \leq 0$ and 1 for $\hat{\beta}_1 > 0$
- Thus:
  - Positive correlation with positive slope.
  - Negative correlation with negative slope.

Test for significant correlation

- Test for significant correlation (linear relationship) between two variables:

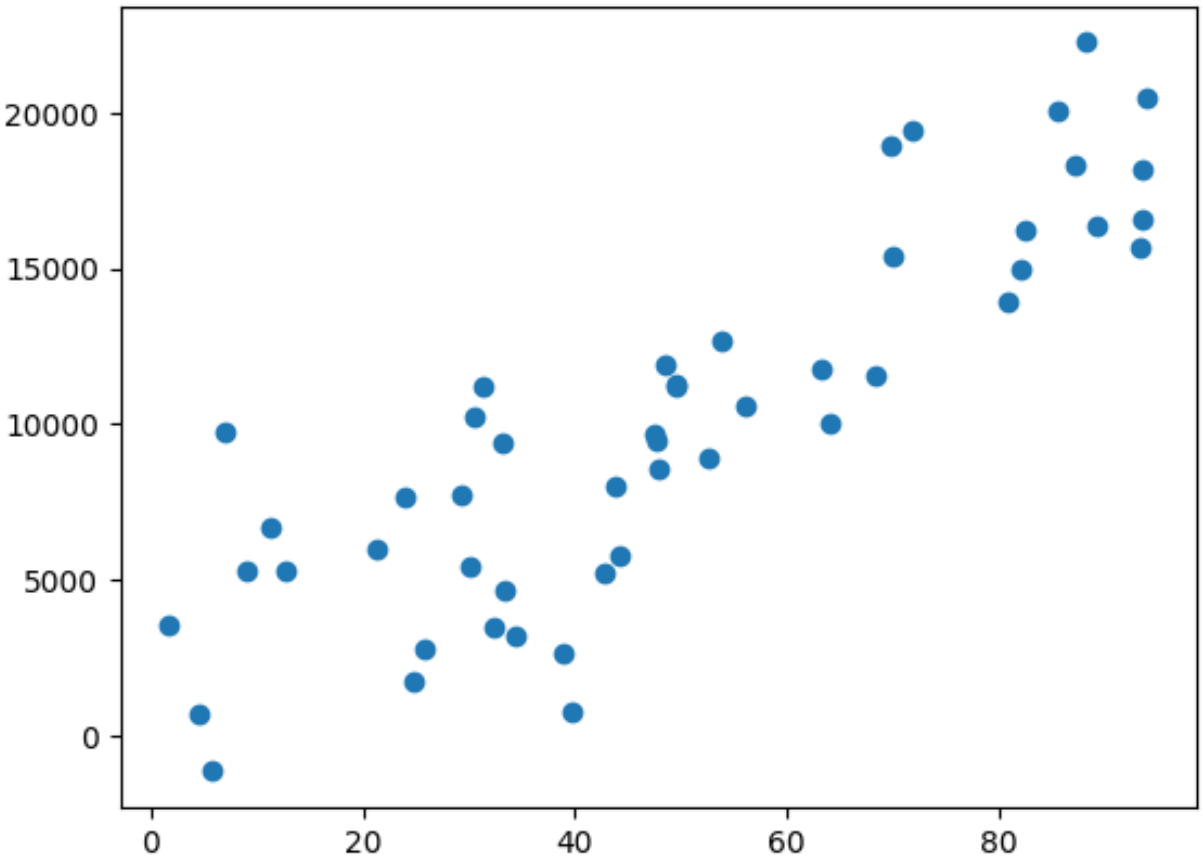
$$H_0 : \rho = 0,$$
$$H_1 : \rho \neq 0,$$

is equivalent to

$$H_0 : \beta_1 = 0,$$
$$H_1 : \beta_1 \neq 0,$$

where β_1 is the slope in the simple linear regression model.

Kahoot



Kahoot!

(x2)

Example: Linear regression with height and weight data

- Go to today's Python notebook in VS Code
  - "Example: Correlation and R^2"



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Residual analysis

Method 5.28

- Check the normality assumption with a QQ-plot.
- Check for any systematic deviations by plotting the residuals (e_i) as a function of the fitted values $(\hat{y}_i)$.

(Method 5.29)

- Is the independence assumption reasonable?

Example: Linear regression with height and weight data

- Go to today's Python notebook in VS Code
  - "Example: Model validation in Python"



Next time

Next time: Multiple Linear Regression

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Checklist

After today, you should be able to:

- Assess from a scatterplot whether it would be relevant to create a simple linear regression model for the data
- Formulate a simple linear regression model and estimate its parameters
- Describe the principle behind "Least Squares Estimation"
- Estimate uncertainties (including confidence intervals) for the parameters in a simple linear regression model
- Perform hypothesis testing to determine whether a parameter is significantly different from zero
- Calculate predicted values for a simple linear regression model, including confidence and prediction intervals
- Describe the relationship between R^2 and the correlation coefficient (and the slope of the regression line)
- Perform model diagnostics for a simple linear regression model (this point is mainly covered in the projects)